

REGINA H REYNOLDS

Molecular biologist turned bioinformatician, with experience translating large-scale omics data into actionable insights for drug discovery. I've led cross-functional analyses across transcriptomics, genomics, and functional screens, collaborating closely across teams to build reproducible pipelines that support impactful R&D decisions. I'm driven by a commitment to scientific rigour, integrity, and transparency — and by a broader desire to make a meaningful contribution to the world through thoughtful, data-driven science.

View this resume online with links at <https://rhreynolds.github.io/cv>



WORK EXPERIENCE

Present
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2025

Senior Principal Scientist, Bioinformatics Lead

AstronauTx

📍 London, UK

2025
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2022

Lead Computational Biologist

CoSyne Therapeutics

📍 London, UK

- Provided scientific and strategic leadership as lead of a 5-person computational biology team, translating high-level objectives into clear milestones and actionable plans, managing tight deadlines and resources, and setting research priorities in collaboration with team members, experimental, engineering, and AI teams, and senior leadership.
- Drove company-wide 'omics initiatives spanning whole-genome sequencing, transcriptomics and CRISPRi screens to support target discovery in glioblastoma multiforme.
- Established scalable, robust, and reproducible computational workflows through adoption of tools such as Nextflow and Seqera Cloud for workflow orchestration, Docker for containerisation, R targets for reproducible exploratory data analysis, and Cruft templates for project scaffolding.
- Contributed to shaping company culture through line management; growing the computational biology team from 2 to 5 people; active involvement in recruitment across multiple teams; and contributing to the development of CoSyne's personal development framework.

2025
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2022

Honorary Senior Research Fellow

University College London

📍 London, UK

- Acted as secondary supervisor to a PhD student and served on the thesis committee for another. Additionally contributed to analyses and provided input on papers across multiple research projects.

2022
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2021

Research Fellow

University College London

📍 London, UK

- Lead analyst involved in processing and analysing transcriptomic data to identify molecular signatures of Parkinson's disease progression. Worked within a large, multi-PI team and coordinated with researchers to align analytical approaches. Work performed using R, Nextflow, and Docker.
- Co-lead of Code and Pipeline Alignment Working Group in the Aligning Sciences Across Parkinson's initiative, facilitating the standardisation of data processing pipelines and coding practices across 10 international research teams, resulting in actionable recommendations to enhance data harmonisation and enable meta-analysis of post-mortem brain tissue datasets.
- Published 1 co-first author research article.

CONTACT

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🐙 [GitHub](#)

in [LinkedIn](#)

📖 [ResearchGate](#)

LANGUAGES

📊 R

🔗 Bash

🐍 Python

TOOLS

🔖 Git/GitHub/GitLab

🔗 Nextflow

🐳 docker

The long-form version of my CV, with a list of conferences attended, teaching experience and voluntary work is available [here](#).

The source code is available [GitHub](#).

Last updated on 2026-03-12.

2016
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2016

Research Assistant

University of Copenhagen

📍 Copenhagen, Denmark

- Led project exploring the interactions between miR-34a, Sirt1 and p53 in a Huntington's disease mouse model, serving as lead experimentalist and analyst; work culminated in a first author publication².
- In addition to these roles, facilitated collaboration with another research group and coordinated experiments with lab technicians, students, and researchers to support project goals.



EDUCATION

2021
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2016

PhD, Bioinformatics

University College London

📍 London, UK

- Thesis: Exploring the importance of cell-type-specific gene expression regulation and splicing in Parkinson's disease³.
- Integrated GWAS summary statistics with bulk/single-cell transcriptomic, eQTL, and chromatin accessibility data to identify cell-type-specific regulatory mechanisms in Parkinson's disease. Methods used included partitioned heritability, COLOC, and MAGMA.
- Applied transcriptomic methods, including cell-type deconvolution, differential splicing analysis, and RNA-binding protein motif analysis, to investigate splicing alterations and their relevance to Lewy body diseases.
- Published 3 first/co-first author research articles and 1 first author review. Successfully secured £10,000 from Signe og Peter Gregersens Mindefond to undertake transcriptional profiling of Parkinson's disease brain tissue.

2016
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2014

MSc, Molecular Biomedicine

University of Copenhagen

📍 Copenhagen, Denmark

- Thesis: Changes in the miR-34a-SIRT1 axis in Huntington's disease
- Grade: A (92.5%)

2013
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2010

BSc, Molecular Biomedicine

University of Copenhagen

📍 Copenhagen, Denmark

- Thesis: Pro-apoptotic factors in Huntington's disease: a study in the R6/2 transgenic mouse model
- Grade: A (96.7%)



KEY PUBLICATIONS

2023

Local genetic correlations exist among neurodegenerative and neuropsychiatric diseases⁴

NPJ Parkinson's disease

- Reynolds, RH, Wagen, AZ, Lona-Durazo, F, Scholz, SW, Shuai, M, Hardy, J, Gagliano Taliun, SA, Ryten, M
- Role: Co-first author, lead analyst and corresponding author.
- Analysis⁵ of local genetic correlations between neurodegenerative and neuropsychiatric disorders, with the aim of identifying genomic regions and genes that may drive pleiotropy.

A full list of publications is available online at <https://rhreynolds.github.io/cv>

- 2021 **Cross-platform transcriptional profiling identifies common and distinct molecular pathologies in Lewy body disorders⁶**
Acta Neuropathologica
- Feleke, R, Reynolds, RH, Smith, A, Tilley, B, Gagliano Taliun, SA, Hardy, J, Matthews, PM, Gentleman, S, Owen, D, Johnson, MR, Srivastava, P, Ryten, M
 - Role: Co-first author and analyst.
 - Transcriptomic analysis⁷ of cell-type-specific changes in the Lewy body diseases.
- 2019 **Informing disease modelling with brain-relevant functional genomic annotations⁸**
Brain
- Reynolds, RH, Hardy, J, Ryten, M, Gagliano Taliun, SA
 - Role: First author.
 - Review of conceptual advances in the generation of brain-relevant functional genomic annotations and among tools that allow integration of these annotations with genome-wide association summary statistics.
- 2019 **Moving beyond neurons: the Role of cell type-specific gene regulation in Parkinson's disease heritability⁹**
NPJ Parkinson's disease
- Reynolds, RH, Botía, JA, Nalls, MA, International Parkinson's Disease Genomic Consortium (IPDGC), System Genomics of Parkinson's Disease (SGPD), Hardy, J, Gagliano Taliun, SA, Ryten, M
 - Role: First author and lead analyst.
 - Analysis of Parkinson's disease common variation, with the aim of identifying cell types and pathways of importance to disease risk.

LINKS

- 1: <https://parkinsonsroadmap.org/research-network/pd-functional-genomics/>
- 2: <https://pubmed.ncbi.nlm.nih.gov/29289683/>
- 3: <https://discovery.ucl.ac.uk/id/eprint/10119171/>
- 4: <https://pubmed.ncbi.nlm.nih.gov/37117178/>
- 5: <https://rhreynolds.github.io/neurodegen-psych-local-corr/>
- 6: <https://pubmed.ncbi.nlm.nih.gov/34309761/>
- 7: <https://rhreynolds.github.io/LBD-seq-bulk-analyses/>
- 8: <https://pubmed.ncbi.nlm.nih.gov/31603214/>
- 9: <https://pubmed.ncbi.nlm.nih.gov/31016231/>